How to Get a Project Topic and Write a Proposal

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Abstract

- This guide is intended to help you produce a final year project. It gives advice on how to gather relevant materials organize it into a suitable from and how to then turn it into a written report. It also describe the convention that should govern the structure of the report.

Introduction

- This guide is meant to help you produce a good final year project report or MSc dissertation. A good report is one that presents your project work *concisely* and effectively. It should contain various materials relevant to the work you have undertaken in respect of your project; it should be organised into a logical framework; and it should be supported by written material that follows well-established academic conventions in a consistent fashion.
- The purpose of the project is, in the context of the degree you are studying, to integrate various aspects of the taught material and to demonstrate your (academic) research skills and your (professional) analysis, design and implementation skills. It gives you the opportunity to conduct in-depth work on a substantial problem to show individual creativity and originality, to apply where appropriate knowledge, skills and techniques taught throughout the degree programme to further oral and written communication skills, and to practise investigative, problem-solving, management and other transferable skills. The management and execution of the project is your responsibility, but you should seek and take advantage of advice from your supervisor.

- When you choose a project, you should do so carefully, to reflect the focus of the degree programme you are enrolled in, your personal interests (the project needs to keep you interested for the whole the academic year) and the ability of the academic staff to support you throughout your project. Projects vary widely in the problem they address and the products they deliver at the end. While the main product of some projects is a piece of software or hardware, other projects produce a systems model or design, and yet others may address some research hypothesis using a theoretical or experimental approach. This means not every project produces a piece of software. In brief, the better defined the problem that your project addresses, the further through the systems lifecycle you should expect to progress in the course of your project. If instead you are addressing a research hypothesis, your main product may be the evaluation of some experiments or a theoretical result.

- So, for example, a project that seeks to develop a logistics planning system for a small business or voluntary organization would be expected to provide a fully operational, fully tested program that meets all the identified needs of the client. However, a project that aims to validate a government policy in a particular area might only achieve the development of a model to confidently simulate the main factors influencing that policy, and identify the research agenda in terms of specifying precisely the data requirements to allow a full investigation of the relevant factors. A scientifically oriented project may focus on the practical or theoretical evaluation of a new rendering approach and compare it with existing approaches, which may involve some implementation, but does not require fully functional software.

- An important point to remember is that the report should describe your work. Large chunks of bookwork describing standard material are unnecessary.

What is a Project Proposal

- The PROJECT PROPOSAL is the background and planning document for the project. Its is an expression of intention of what study/work you wish to carry out. It must be done with professional care and thoroughness.

Project Proposal Outline

- Title Page
- Table of Contents
- Introduction and Background
- Problem Statement
- *Previous Work
- Scope and Context of the Study
- Aim and Objectives of the Study
- Overview of Methodology
- Significance of the Study
- Timeline
- References

Statement of the Problem

- Give a brief, general statement of the problem to be investigated or solved by the project.
- Assume the reader has little knowledge of the subject.

- Previous Work
- This is an historical or conceptual survey of relevant work done in the area by
- previous investigators. Each contribution must be accompanied by
- appropriate references to be listed in the reference section.

Aim:

- Example:
- The aim of this project is to develop software for the improved planning of the routing of delivery vehicles to customer locations, that reflects the forecast availability of each customer to receive goods.
- *Research question(s)*:
- In order to demonstrate the achievement of the stated aim, this project will identify route planning software currently in use and the underpinning algorithms, define appropriate performance metrics, determine how to express constraints on an alternative algorithm, develop an improved algorithm and demonstrate on what basis it is judged an improvement, and implement the improved algorithm in a usable and robust software package.
- **Example 2:**
- *Aim:*
- The aim of this project is to develop a business strategy for organisation X that will improve the survivability of X in the face of increasing global competition.
- *Research question(s)*:
- In order to develop a business strategy it will be necessary to identify key stakeholders and determine their vision for the organization at the end of the strategic planning timeframe, assess the likely outcome, in terms of the organization's survivability, of maintaining the current strategy, and develop and assess an alternative set of activities to achieve the stated vision.

How to get a project topic

- You will work with your academic and workplace project supervisors to identify and select a suitable project. The project must be agreed by both supervisors before work can begin. Before a project is signed off, it must be clear that it:
- Can be completed within the stated time constraints.
- Can be completed within extant financial constraints.
- Can be completed given available technical resources (e.g. tools and network access).
- Is at an appropriate level of difficulty (not too challenging and not insufficiently challenging).
- Covers, as far as possible, the full range of skills and knowledge described in the apprenticeship standard (e.g. scoping, planning, costing, analysis, design, implementation, testing, etc.).
- Is original in nature (e.g. has not been done before).
- Satisfies all ethical and legal requirements for software engineering projects (e.g. copyright, confidentiality, etc.).

Content: Introduction

A good introduction should tell the reader what the project is about without assuming special knowledge and without introducing any specific material that might obscure the overview. It should anticipate and combine main points described in more detail in the rest of the project report. Also, importantly, it should enthuse the reader about the project, to encourage them to read the whole report. Normally it should include such things as:

- the aim(s) or goal(s) of the project;
- the intended audience or "beneficiaries" of the work done;
- the scope of the project;
- the approach used in carrying out the project;

- assumptions on which the work is based; and
- a broad summary of important outcomes.